

ABHINAV KAUSHIK

Senior Software Engineer

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SUMMARY

Senior Software Engineer with 8+ years designing and shipping production backend services for ML and AI systems across Healthcare, Telecom and AIOps, with an MSc in Machine Learning and AI from Liverpool John Moores University. Builds and maintains Python microservices and APIs with FastAPI, Docker and Kubernetes on GCP, managing MLOps pipelines with MLflow and supporting data scientists from prototype to production. Works across the full stack, from LangGraph multi-agent pipelines and SLA-managed prediction services to React and Next.js front-ends, with automated evaluation, tracing and observability, translating business requirements into measurable outcomes with stakeholders. Looking for opportunities in Berlin and willing to relocate. Currently building ThinkTree.AI, a non-profit AI-powered learning platform designed to support students with ADHD and context-switching challenges.

PROFESSIONAL EXPERIENCE

Senior Software Engineer | United Health Group | Bengaluru, India Nov 2025 – Present

- Developed 21+ automated HTML, TypeScript and Next.js mockup generators on a Node.js stack, turning Jira tickets and PRD requirements into working React prototypes and carrying them from POC through to production.
- Deployed a production Python backend service for lab document extraction with Claude Opus 4.7 and Amazon Textract, integrated with LIMS via REST API, processing 1,800 lab documents and automating 800 Jira tickets.
- Engineered a production multi-agent analytics backend in LangGraph and PySpark for AARP medical plans, maintained under SLA, answering business questions with multi-turn reasoning and live charts, cutting manual analysis by 98%.
- Built automated evaluation and observability pipelines with LLM evaluators, tracing and failure detection for Question Answering and Summary agents, reducing human-in-the-loop review by 43%.
- Fine-tuned a cross-encoder on positive and negative pairs to re-rank dense retrieval over AARP survey data, improving retrieval accuracy and shipping it as a versioned MLflow model artifact.

Senior Machine Learning Engineer | Viavi Solutions | Chennai, India Sep 2024 – Nov 2025

- Developed an internal RAG-based Project Discovery microservice in LangGraph, combining semantic retrieval with LLM reasoning; deployed on GCP with Docker and Kubernetes, reaching a 90% match rate against existing solutions and used by data scientists across the organization.
- Built an event-correlation engine clustering alarms with Sentence-BERT and DBSCAN, packaged as a Python backend service, to trace cascading network alarm floods back to the root event, improving alarm classification by 73.5%.
- Mentored the team on service architecture, clean code, unit testing with pytest and production monitoring, while prototyping 15+ AI applications for AIOps clients, with 7 moving to production or paid engagements.

Data Scientist | Viavi Solutions | Bengaluru, India Nov 2021 – Sep 2024

- Deployed a production SLA-breach prediction service in Python, FastAPI and MLflow with Kubernetes orchestration over TimescaleDB (PostgreSQL) time-series data, using GPT-3.5 for real-time root-cause analysis per breach, reducing network downtime by 67%.
- Shipped anomaly detection and demand forecasting on 15+ KPIs across 500+ telecom network devices using Isolation Forest and Holt-Winters, enabling earlier detection and proactive network planning.
- Delivered technical POCs and consulting for 15+ clients across AIOps and ML platforms, contributing to an 11% increase in POC and mentoring revenue.

Data Analyst | Optum | Gurugram, India Jun 2021 – Nov 2021

- Implemented a reranker for an existing recommendation service, optimizing renewal suggestions and increasing conversion by 2%.
- Developed a Python and Flask parser automating extraction and transformation of raw data into structured datasets feeding downstream analytics and reporting.

Data Engineer | Nippon Data Systems | Gurugram, India Nov 2019 – May 2021

- Engineered SQL and PySpark data pipelines automating BAU medical-audit reporting on large healthcare datasets, cutting manual reporting effort by 95%.
- Built real-time log observability for the COVID QRT team with Elasticsearch and Kibana, improving traceability efficiency by 80%.

Associate Consultant | Saksoft | Gurugram, India Dec 2017 – Nov 2019

- Developed an XGBoost propensity model predicting personal-loan applications within 7 days, enabling targeted outreach and customer prioritization.
- Built customer dashboards for financial and product insights with QlikSense, SQL and SAP HANA, increasing user engagement by 40%.

PROJECTS

Think Tree – ML Study Platform

- Built a non-profit AI-powered ML study platform for people with ADHD and context-switching challenges, designed to help users organize learning and maintain study context.

Ad Creative Performance Scoring Service

- Built a Python/FastAPI microservice scoring ad creative and campaign KPI signals with a LightGBM model and time-series features, deployed on GCP with Docker and Kubernetes, MLflow experiment tracking, and automated performance monitoring – directly applicable to creative-optimization and campaign-performance workflows.

RFP Multi-Agent Proposal Automation Platform

- Built a multi-agent RFP platform where an LLM supervisor orchestrates intent detection, retrieval, proposal generation, document matching and market research, with RAG over historical proposals.

Financial Fraud Detection – Mar 2025

- Built a fraud detection and reasoning workflow on the JPMorgan Synthetic Dataset, pairing LightGBM scoring with LLM-generated explanations for interpretable financial-risk analysis.

EDUCATION

Master of Science, Machine Learning and AI Liverpool John Moores University	Jan 2021 – Jan 2023
Bachelor of Technology, Information Technology Jamia Hamdard	Aug 2013 – Aug 2017

TECHNICAL SKILLS

AI / GenAI: Generative AI, LLM, RAG, Agentic RAG, Agentic AI, LangGraph, LangChain, Claude, OpenAI, Embeddings, Cross-Encoder, Fine-Tuning, Hugging Face Transformers, NLP, Tool Orchestration, Amazon Textract

Machine Learning: PyTorch, Scikit-learn, XGBoost, LightGBM, Random Forest, Isolation Forest, LSTM, SBERT, DBSCAN, Time Series Forecasting, Demand Forecasting, Anomaly Detection, Statistical Modeling, Regression, Deep Learning

Evals / Observability: LLM Evals, LLM Evaluators, Agent Tracing, Observability, Elasticsearch, Kibana, Log Analysis

Programming / Data: Python, SQL, PySpark, Pandas, NumPy, FastAPI, Flask, Hive, Spark SQL, REST API, Pydantic, Unix

Frontend / Full-Stack: React, Next.js, TypeScript, JavaScript, Node.js, HTML, CSS, Microservices

MLOps / Cloud: MLflow, Docker, Kubernetes, Airflow, DVC, Terraform, Jenkins, Bamboo, Bitbucket, GCP, Azure, Vercel

Engineering Practices: Unit Testing, pytest, Code Review, CI/CD, Monitoring, Observability, Service Architecture, Microservices, Mentoring

Data Platforms: Databricks, Databricks SQL, Apache Spark, Supabase

Databases: PostgreSQL, TimescaleDB, MySQL, Vector DB, pgvector, ArangoDB, Cosmos DB

Search: Apache Solr, Lucene Indexing, Information Retrieval

BI: Power BI, QlikSense, SAP HANA

LANGUAGES

English: Native

German: A1 – Pursuing

ADDITIONAL INFORMATION

- Network Intern, Airport Authority of India
- Hadoop Intern, Ducat Technologies